

DRAFT (for review; not for deposit)

A Blind Benchmark of Decipherment Methods on Synthetic Scripts Calibrated to Indus Statistics

Arnav Goel, Woodstock School, Mussoorie, India

arnavgoel@woodstock.ac.in

Abstract

Arguments about the undeciphered Indus script often rest on methods never tested on corpora of Indus size and shape with a known answer. We write five known languages and several non-linguistic sign systems in invented scripts, calibrate them to published Indus statistics (2,906 texts, 4.6 signs per text), hide the answer keys, and test eight frozen method families at three levels of solver knowledge. Four results replicate on fresh seeds. Entropy statistics do not separate synthetic languages from an i.i.d. random control at Indus scale. Without a related language among the candidates, mean recovery of sign values stays at or below 4.8% in every run. At equal total size, longer inscriptions help more than more inscriptions, in both text samplers and in logo-syllabic corpora. And published Indus statistics do not fix how decipherable an Indus-scale corpus is: two samplers that the published statistics cannot tell apart differ roughly three- to fivefold when a related language is available, and for logo-syllabic corpora only when it is the sole reference. A fifth, archaeological result is a prediction, not a measurement. The main solver matches the best published accuracy on English substitution ciphers. All limitations, including an optimistic synthetic relative, few source languages and post-hoc decisions, are listed.

Plain-language summary

[AUTHOR REWRITE. Current draft text, kept for reference:]

The Indus script, used in South Asia about 4,500 years ago, has never been read. Its inscriptions are very short, and no text repeats an Indus message in a readable script. We asked: could today's computer methods read writing like this if the answer were known? We wrote five known languages in invented scripts, matched them to published Indus statistics, and hid the key. The main method, expectation-maximisation (EM), guesses what each sign stands for, checks how well the guesses fit a known language, and improves them. We score token accuracy: the share of all signs in the texts that a method reads correctly. Methods were given either a close relative of the hidden language (related), a relative mixed with unrelated languages (candidates), or unrelated languages only (none). Without a relative, methods read about 5% of signs or fewer on average. With one, results depended strongly on corpus properties that published statistics leave open.

1. Introduction

[AUTHOR REWRITE: the first two paragraphs. Current draft text, kept for reference:]

The Indus inscriptions are short. The concordance of Mahadevan (1977) holds 2,906 texts with 13,372 sign occurrences, about 4.6 signs per text (Yadav et al. 2010; Rao 2018), and the longest text is commonly given as 17 signs on a single surface (Farmer, Sproat & Witzel 2004). Published sign lists count between 386 and about 700 sign types, depending on how variants are grouped (Parpola 1994 and Wells 2015, both via Rao 2018; Mahadevan 1977; Wells 2006 via Yadav et al. 2010). No bilingual text is known, and no language of the Harappan civilisation is agreed.

Two kinds of argument have been made about this corpus. Some assign readings to the signs. Others use corpus statistics to argue that the signs do, or do not, encode language: conditional entropy (Rao et al. 2009), n-gram models (Yadav et al. 2010), and, on the other side, text brevity (Farmer, Sproat & Witzel 2004) and the finding that entropy-type measures do not separate writing from non-linguistic symbol systems (Sproat 2010, 2014). Both kinds of argument depend on a further question: would the method work on a corpus of Indus size and shape whose answer is known?

[END AUTHOR REWRITE]

We test that question directly. We write known languages, and sign systems that encode no language, in invented scripts, calibrate each corpus to published Indus statistics, hide its answer key, and measure what existing methods recover. The result is a statement about methods on Indus-like data. This work does not decipher the Indus script or say anything about what language it records, or whether it records a language at all. Nothing in it bears on what the signs mean.

Two scope statements apply throughout. "Decipherment" here means substitution-style recovery of sign values only, not grammar, meaning or the identity of an unknown language. And the benchmark languages (Sanskrit, Old Tamil, Sumerian, Latin, Finnish) were chosen for licences and for spread across families and script types; they imply nothing about the language of the Indus inscriptions, and how well any of them is recovered says nothing about Harappan.

Defining "Indus-like" turned out to be the hard part. Published statistics fix the number of texts, their length, the sign inventory and several frequency and positional measures. They do not fix how often texts repeat, how many sign types are variants of one another, or how text is sampled. We find that choices these statistics leave open, here the way text is sampled, change recovery roughly three- to fivefold when a related language is available, while duplicate rate and inventory within one sampler matter less under the frozen rule. So we report a map, not a single Indus number.

Findings.

1. Entropy statistics do not separate synthetic languages from an i.i.d. random control (independent, identically distributed signs) at Indus scale.
2. Without a related language among the candidates, mean recovery stays at or below 4.8% in every run. No corpus reaches 50% of tokens; the best reaches 28%.
3. Under the frozen rule, longer inscriptions help more than more inscriptions in both samplers, logo-syllabic corpora included (Table 6).
4. Published Indus statistics do not fix how decipherable an Indus-scale corpus is. Two samplers that the published statistics cannot tell apart differ roughly three- to fivefold when a related language is available (Table 7). For logo-syllabic corpora,

the most Indus-relevant type, this contrast appears only when the related language is the sole reference; among candidate languages it is unresolved.

5. Archaeologically defined subsets of the corpus are predicted to sit at different points on the map. These are predictions, not measurements.

The paper contributes a calibrated script generator with every target cited and flagged, eight frozen method families behind one interface, a protocol scoring four tasks separately (language-level intervals, three calibration regimes, three knowledge tiers, fresh-seed replication), a sensitivity map with published reference points, and a blind challenge with hash-committed keys.

2. Related work

Structural statistics. Rao et al. (2009) compared conditional entropy across Indus and reference systems and argued for linguistic structure. Yadav et al. (2010) fitted n-gram models and reported the statistics behind most of our calibration targets. Farmer, Sproat & Witzel (2004) argued from brevity and repetition that the signs were not writing, and Sproat (2010, 2014) showed that entropy-type statistics do not reliably separate writing from non-linguistic symbol systems. Lee, Jonathan & Ziman (2010) proposed an entropy-based classifier for Pictish symbols, which we include as a method.

Closest prior work. Nair (2026), a preprint, tests 1,916 deduplicated ICIT inscriptions against heraldic and administrative baseline generators and seven attested non-linguistic corpora, on four properties from the Farmer-Sproat-Witzel critique, and places the corpus between the baselines. Tiwari (2026) analyses 6,579 inscriptions with sign-form clustering, entropy, Kullback-Leibler divergence and a BiLSTM, and reports directional asymmetry and structured combinatorial patterns. Both ask what kind of sign system the real corpus is. We never touch the real corpus; we score methods against hidden keys on synthetic corpora. Nair's ICIT duplicate rate is used only as a reference point.

Computational decipherment. Knight et al. (2006) framed decipherment as unsupervised learning of a substitution model with expectation-maximisation (EM); Berg-Kirkpatrick & Klein (2013) showed that many random restarts help. Snyder, Barzilay & Knight (2010) recovered Ugaritic with Hebrew as a known relative, and Luo, Cao & Barzilay (2019) recovered Ugaritic and Linear B cognates with a neural model and minimum-cost flow. Each success had a close relative and far more text per document than the Indus corpus, which is why our knowledge tiers keep these conditions apart. Our EM cognate-matcher borrows only Luo et al.'s one-to-one assignment idea; it contains no neural network and says nothing about neural methods.

Script type and segmentation. We estimate script type from inventory size with a Chao (1984) estimate, and segment with branching entropy (Harris 1955; Tanaka-Ishii 2005).

Archaeological context. Kenoyer & Meadow (2010) and Kenoyer (2020a, 2020b) describe how inscribed objects at Harappa differ by type and period: seals are almost all unique, tablets often occur as copies or same-mold duplicates, and the script changed over roughly 700 years. We use these statements only qualitatively (Section 4.5).

3. Benchmark design

3.1 Sources and scripts

Plaintext comes from five languages in four families: Sanskrit (Rāmāyaṇa, Digital Corpus of Sanskrit, CC BY 4.0), Old Tamil (17 Sangam root texts, Project Madurai), Sumerian (CDLI bulk ATF dump, with Ur III seal inscriptions), Latin (Caesar and Vergil) and Finnish (Kalevala), the last two from Project Gutenberg. CDLI allows reuse "according to common and fair academic practice" with citation, which is not an open licence, so Sumerian material stays out of published challenge rounds. No source text is redistributed. Each source is cut into short, seal-like clauses: names, epithets, titles, formulae and, for Sumerian, real seal inscriptions.

3.2 Synthetic scripts

Words are written alphabetically (phonemes), syllabically (CV, V and C units), logographically (word forms) or logo-syllabically (the 250 most frequent words as logograms, the rest spelled syllabically). Units map to random sign identifiers, with optional allographs, homophones, polyvalent signs, determinatives, word dividers and reading direction.

3.3 Calibration targets

Each target in `config/indus_targets.yaml` carries its citation, tolerance, reference corpus size and verification status (Table 1). Size-dependent statistics are computed on subsamples of the reference size.

Table 1. Calibration targets.

Statistic	Target	Source
Texts	2,906	Mahadevan (1977), via Yadav et al. (2010) and Rao (2018)
Mean length	4.60 signs	Mahadevan (1977), via Yadav et al. (2010) and Rao (2018)
Longest text	commonly given as 17 signs on a single surface	Farmer, Sproat & Witzel (2004)
Sign inventory	400-700	published sign lists (Section 1)
Signs covering 80% of tokens	69	Yadav et al. (2010)
Share of the most frequent sign	0.10	Yadav et al. (2010)
Text-final signs covering 80% of finals	23	Yadav et al. (2010)
Text-initial signs covering 80% of initials	82	Yadav et al. (2010)
Median length	4	derived from Yadav et al. (2010), Fig. 2 (2,591 texts, mean 3.92, max 14; not like-for-like with 2,906 texts at 4.6 signs); originally unsourced (Appendix A)
Hapax share of sign types	0.27	Possehl (2002), read via a secondary source

The sampler also uses the median to set the spread of text lengths.

The duplicate-text rate is not a target, because no source fixes it for the whole corpus. Two published values serve as reference points. For Mahadevan's concordance we read

0.354 from the vector data of Yadav et al. (2010, Fig. 2), or 0.281 without its four most repeated texts. For the ICIT corpus an unverified preprint gives 0.237 (Nair 2026). The Yadav et al. figure plots 2,591 texts, not 2,906, without saying which are left out.

3.4 Calibration regimes

`full` imposes every target. `holdout` imposes only size, length and inventory, so that no Task A method is rewarded for frequency or positional statistics we injected. `wrong_prior` imposes deliberately wrong targets. Results lead with `holdout` wherever calibration could be exploited.

3.5 Non-linguistic controls

The controls encode no language, and they are of two different kinds. Statistical baselines have no meaningful structure: the i.i.d. random control and the rigid control of Rao et al. (2009), and an adversarial Markov chain tuned to the entropy ratio of the language corpora. Structured non-linguistic systems are meaningful and structured but are not language: heraldic bearings built with the rule of tincture (one sign per visual element), administrative slot tags and sparse Markov emblems. Rao et al. call the i.i.d. random control type 1 and the rigid one type 2; our internal keys number them the other way round, so we name them by behaviour. Japanese kamon descriptions are text and are reported separately as a contaminated control.

3.6 Knowledge tiers and the synthetic sister language

Tasks C (language family) and D (sign values) run at three levels of solver knowledge. `related` gives one reference language, a synthetic sister of the hidden language. `candidates` adds every other benchmark language. `none` removes the hidden language and its sister. Only `candidates` and `none` matter for the Indus case: no close relative of the Harappan language is agreed, and we do not know which of the two applies.

The sister is built from the half of the source text not used for the hidden corpus: a seeded 30% of consonants and of vowels are permuted by a regular, bijective correspondence, and 20% of word types are replaced by unrelated words of the same length. **It is therefore much closer to the hidden language than any attested relative of an undeciphered script is likely to be**, in three ways (Table 2).

Oracle sound correspondences. The solver's frozen cognate step maps sister units to the hidden language with the true table. We score before this step and report its gain only in tables.

Shared text. The original sister split clauses between the halves by position, so a repeated clause could land in both. The corrected sister splits by a hash of clause content, so no clause occurs in both halves; the fix moved no Indus-point mean outside its earlier interval. Word-for-word overlap stays high anyway, because short phrases recur throughout a text and no clause split can remove them. A real relative, with its own vocabulary, morphology and word order, would share far fewer. Every sister-tier result carries this caveat.

No structural divergence. Word order and morphology are untouched; only phonemes and a fifth of the vocabulary change. Distance is a free parameter, swept over four settings in the appendix.

Table 2. How much closer the synthetic sister is than a real relative (Indus point). Overlap is measured over 12 corpora per language (4 script types, 3 seeds).

Property	Sampler A	Sampler B
Gain from the oracle cognate step, related tier	7-8 points	17-19 points
Largest gain on a single corpus	up to 51 points (either sampler)	
Hidden texts found verbatim in the sister's half, corrected sister, range of language means	30-53%	43-56%
... Sanskrit / Old Tamil / Sumerian / Latin / Finnish	46 / 36 / 30 / 36 / 53%	56 / 46 / 55 / 43 / 48%
... range over single corpora	8-90% (both samplers)	
Stricter sister (Section 4.6): sister clauses removed	33%	42%
Hidden-text word pairs still in the sister: corrected / stricter	31% / 8%	35% / 7%

Section 4.6 repeats the key comparisons with a stricter sister that drops every clause containing any hidden text's word sequence.

3.7 Methods

Eight method families run on every corpus behind one interface: conditional and block entropy (after Rao et al. 2009), a bigram Markov model (after Yadav et al. 2010), positional histograms, segmentation, script-type estimation by rule and by classifier, a two-parameter tree (Lee, Jonathan & Ziman 2010), a multi-feature language classifier, Knight-style EM, an EM cognate-matcher and a frequency-rank baseline. Method code was frozen before the final runs (Section 6); later changes touch only data generation and analysis.

EM needs a rule for choosing among reference languages. The **original rule**, written before the runs, selects by raw likelihood with learned emissions for a pooled rare-unit state. When it chose Sumerian for almost every corpus, we wrote a **revised rule**: uniform emissions for that state, and selection by bigram gain normalised by the reference's own mutual information. The original rule is primary throughout; the revised rule is post hoc.

3.8 Tasks and scoring

Task A asks whether a corpus encodes language (balanced accuracy; chance 0.5), Task B its script type (accuracy; chance 0.25), Task C its language family (chance one over the number of candidate families), and Task D the value of each sign, scored as the share of sign tokens read correctly; "success" means at least 50%. Rules for Tasks A and B are learned leave-one-source-out. Means carry 95% cluster-bootstrap intervals that resample source languages (or control families) and then corpora within them, because corpora from one source are not independent. As a check that the Task D scorer is not generous, permuting a solver's predicted values across signs drops a 93% corpus to 0.9%. Positive controls against published results are in Section 4.9.

Corpora use seeds 0-2, replicated on fresh seeds 3-5, and two text samplers. Sampler B differs from A only in drawing the opening word type first; it was meant to fix a missed text-beginner target, and did not. Both meet the size and length targets in every corpus and the others at similar, not identical, rates. At the Indus point all targets are met by

13% of sampler-A and 3-7% of sampler-B language corpora (corrected-sister runs; full table in the appendix).

3.9 Sensitivity sweep

Duplicate-text rate and sign inventory are set exactly and varied independently on a 6 by 5 grid at the Indus point (duplicates 0-0.5, inventory 400-800). Duplicates come from assembling a corpus out of distinct texts plus popularity-weighted copies; inventory comes from the allograph rate alone, set by bisection. Because no tested method merges allographs, the inventory axis measures robustness to unmerged variants, not to a larger sign system, and every map says so. A cell counts only if its corpora hit both settings without distorting text lengths.

The plausible box (duplicates 0.2-0.4, inventory 400-700) brackets the published duplicate rates and sign-list sizes. It entered the run profile after the sweep had started, after a test rendering of the first 24 records and a count of reachable cells, but before recovery was tabulated, and was first committed together with the results (Appendix A). Published values are marked on the map as reference points, not targets.

4. Results

Table 3. Score labels. Every number in this section is primary unless a table row says otherwise. Intervals are 95% and resample whole source languages.

Label	Meaning
Primary	Original (frozen) EM rule, scored before the cognate step, corrected sister language
With oracle sound correspondences	The same runs, scored after the cognate step
Upper bound	Original sister language (known overlap), scored after the cognate step; runs made before the sister split was corrected
Revised rule	The EM selection rule written after the original rule had run; post hoc

The sensitivity sweep ran with both sister languages; the corrected one supplies Sections 4.4-4.5.

4.1 Entropy statistics fail at Indus scale

The entropy result is the clearest, and it involves no reference language. At the Indus point the plug-in ratio of conditional to unigram entropy, over the full sign alphabet, is 0.470 for an i.i.d. random control (420 signs) and 0.477 for the synthetic languages (Table 4, Figure 1). At this size the statistic cannot tell synthetic languages apart from an i.i.d. random control. After Rao's merge to the 100 most frequent signs, the i.i.d. random control even scores higher. Structured non-linguistic systems are not random, and are tested separately.

Table 4. Entropy ratio $H(X_2|X_1)/H(X_1)$ at the Indus point.

Statistic	i.i.d. random control	Synthetic languages
Full sign set, seeds 0-2	0.470	0.477 (10th-90th percentile 0.35-0.67)
Full sign set, fresh seeds	0.467	0.476
Rao's merge to the top 100 signs	0.696	0.605

A classifier built on the statistic reaches balanced accuracy 0.77 [0.57, 0.90] when the sampler is calibrated to every target (full). Under holdout it falls to 0.57 [0.32, 0.81], and on fresh seeds to 0.49 [0.25, 0.75]; both include chance. Under full, every Task A method labels at least one structurally non-linguistic control family as language in all its corpora, the rigid control in the entropy classifier's case. Language detection at Indus scale is unresolved, and the full score is a product of calibration.

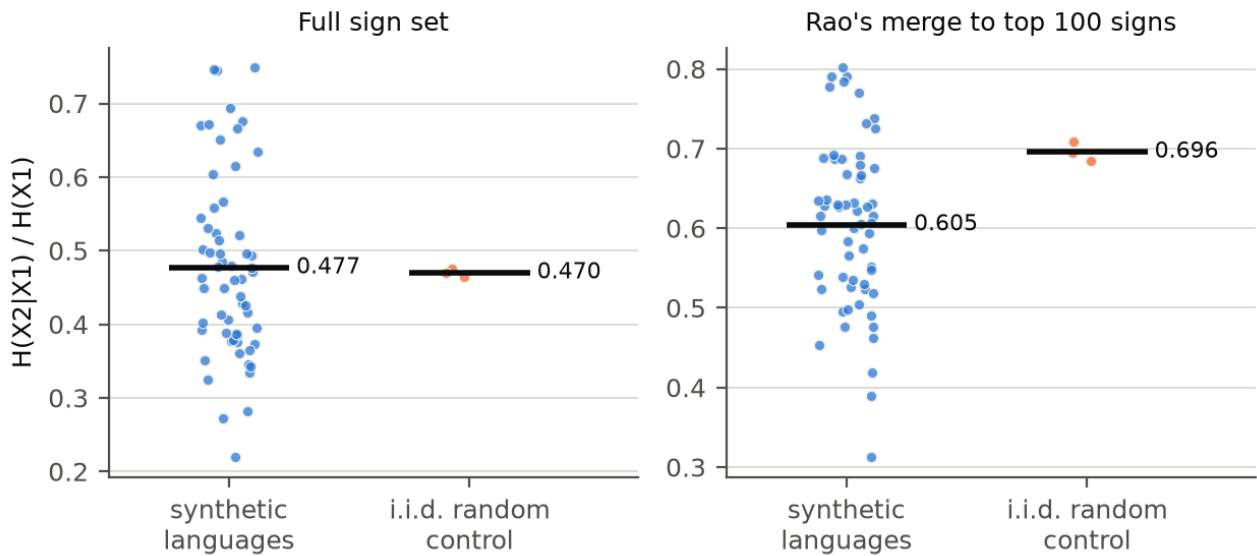


Figure 1. Entropy ratio $H(X2|X1)/H(X1)$ at the Indus point (2,906 texts, mean 4.6 signs; full calibration, seeds 0-2): full sign set (left) and Rao's merge to the 100 most frequent signs (right). Dots are corpora (60 synthetic-language corpora, 3 i.i.d. random-control corpora); bars are means. Score label: not applicable, since Task A uses no reference language.

4.2 Without a related language

The hidden language and its sister are excluded here, so the sister's problems cannot reach this result. Mean recovery stays at or below 4.8% in every run, no corpus reaches 50% of tokens, and the best reaches 28% (Table 5).

Table 5. Recovery in the none tier.

Run	Mean, original rule	Mean, revised rule	Best single corpus (original / revised)	Corpora $\geq$ 50%
Sampler A, seeds 0-2	1.8%	3.3%	16.4% / 18.1%	0 of 60
Sampler A, seeds 3-5	2.4%	3.0%	19.7% / 23.8%	0 of 60
Sampler B, seeds 0-2	3.3%	4.8%	23.2% / 27.6%	0 of 60
Sweep, original-sister hidden split, all valid cells	at most 4.6% per cell	at most 7.1% per cell	24.6% / 27.9%	0 of 1,571
Sweep, corrected-sister hidden split, all valid cells	at most 4.6% per cell	at most 6.3% per cell	26.5% (either rule)	0 of 1,451

Inside the plausible box the mean is at most 4.3% (fixed-composition panel: the same 10 language-script combinations in every box cell), and it is never above 7.1% anywhere on the grid. The corrected-sister sweep gives 4.3% and 6.3%. At 20 signs per text the tier still stays at or below 6.5%.

4.3 Longer inscriptions vs more inscriptions

Length beats volume. Under the frozen rule, 2,906 texts of 10 signs yield more than 6,317 texts of 4.6 signs, about 29,000 tokens each, in both samplers (Table 6, Figure 2). The advantage holds from 10 signs per text in sampler A and from 6 in sampler B, and for logo-syllabic corpora alone. Without a relative both conditions stay at or below 6.5%, with a small edge for length.

Table 6. Longer vs more inscriptions at equal total tokens, candidates tier.

Sampler	Score	Longer	More	Difference, paired [95% CI]
A	Primary	18.0%	2.6%	15.4 [5.8, 24.9]
B	Primary	35.6%	18.3%	17.4 [8.5, 25.0]
A	Primary, fresh seeds (size interpolated on log tokens)	17.7%	3.4%	no interval
B	Primary, fresh seeds (size interpolated on log tokens)	32.2%	11.8%	no interval
A	Primary, logo-syllabic corpora only	12.7%	0.2%	
B	Primary, logo-syllabic corpora only	27.8%	6.4%	
A	With oracle sound correspondences	28.6%	2.7%	25.9 [10.5, 41.4]
B	With oracle sound correspondences	54.9%	27.6%	27.4 [12.8, 39.0]

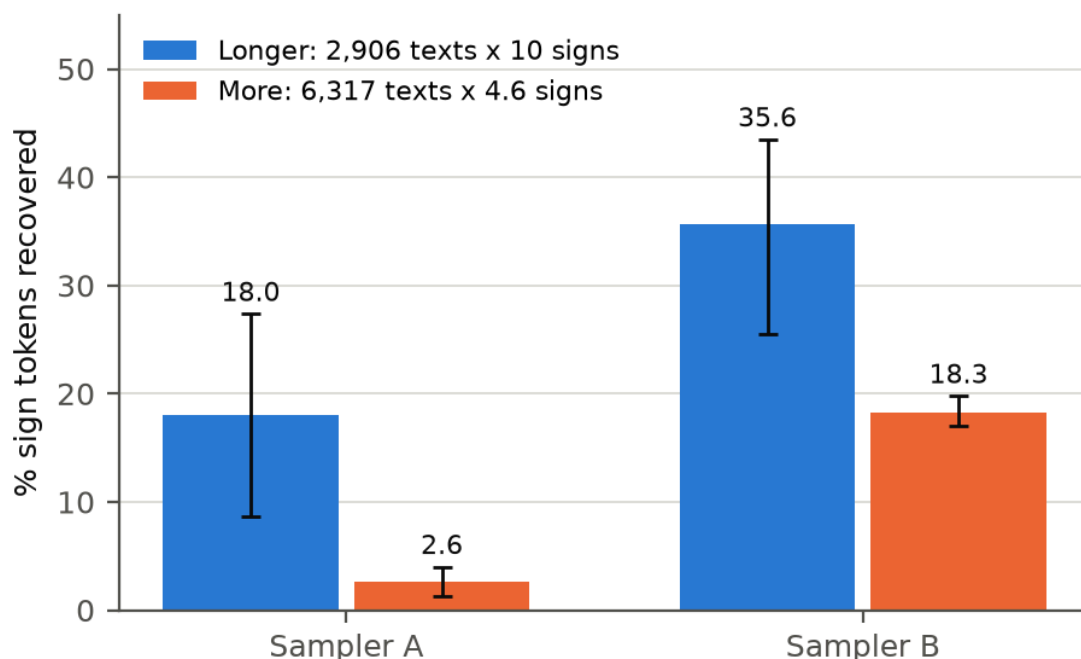


Figure 2. Longer vs more inscriptions at equal total tokens (about 29,000), candidates tier, original rule, both samplers. Score label: primary. Bars are means over 60 corpora; whiskers are 95% intervals resampling whole source languages.

4.4 Difficulty is not fixed by published Indus statistics

Published Indus statistics do not fix how decipherable an Indus-scale corpus is. Two samplers that the published statistics cannot tell apart (neither meets every target in more than 13% of language corpora) differ roughly three- to fivefold when a related language is available. For logo-syllabic corpora, the most Indus-relevant type, this contrast appears only when the related language is the sole reference; among candidate languages it is unresolved.

Table 7. Sampler contrast, candidates tier, Indus point.

Run	Sampler A	Sampler B	Ratio
Primary, seeds 0-2	2.4% [1.2, 3.8]	12.9% [7.8, 17.4]	5.3x
Primary, seeds 3-5	2.7% [1.2, 3.8]	9.4% [4.7, 14.0]	3.5x
Primary, stricter sister (Section 4.6)	2.6% [1.2, 4.2]	9.3% [5.4, 13.5]	3.7x
Revised rule (post hoc), seeds 0-2	9.1%	25.1%	
With oracle sound correspondences, seeds 0-2	2.8%	18.9%	
Upper bound, seeds 0-2	2.1%	15.0%	

No sampler-A corpus reaches 50% of tokens in either seed set; 6 of 120 sampler-B corpora do. The logo-syllabic exception has a simple cause, given in Section 4.7. In the related tier the contrast is smaller in ratio but larger in points: 15.9% against 30.2%.

The samplers differ only in how text windows are drawn, and both meet the size and length targets in every corpus. They also respond differently to size: from 500 to 50,000 texts, sampler A rises only from 2.5% to 3.7%. Sampler B rises from 3.5% to 18.7%.

Within one sampler, duplicate rate and inventory matter much less (Table 8, Figure 3). Across the plausible box, sampler A stays between 2.9% and 4.2% in the box panel (the 10 combinations valid in every box cell). The box does not bound recovery across samplers, though. The 37 sampler-B corpora whose measured duplicate rate and inventory fall inside it (seeds 0-5) average 11.9%, and the best reaches 72.0%.

Table 8. Sampler A across the plausible box, candidates tier.

Quantity	Primary	Upper bound, with oracle sound correspondences
Range of cell means (box panel)	2.9-4.2%	3.5-5.1%
Median within-combination spread	2.1 points	2.1 points
Largest within-combination spread	7.5 points	7.6 points
Revised rule (post hoc): range of cell means	5.6-14.2%	8.7-21.8%
Revised rule: largest within-combination swing	44 points	62 points

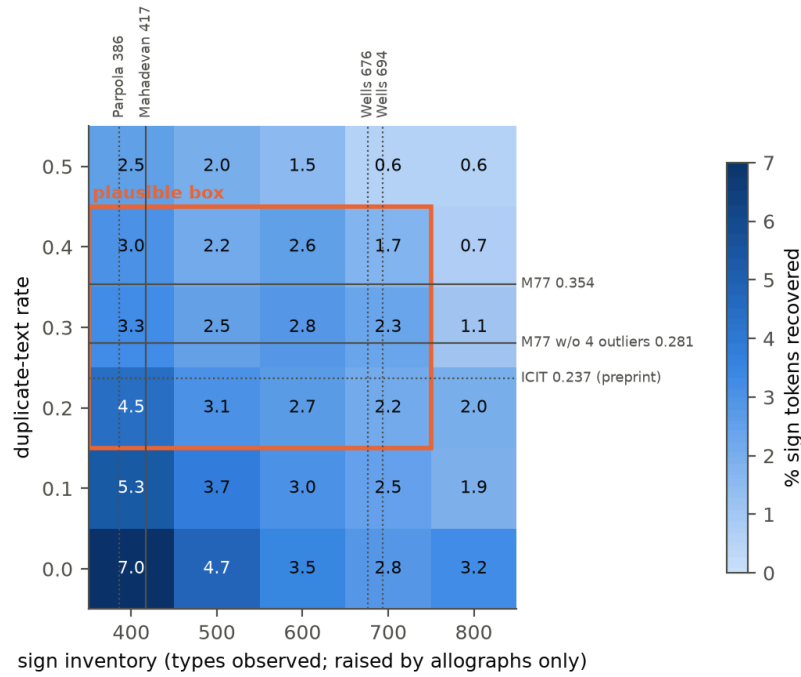


Figure 3. Sensitivity map, sampler A, candidates tier; original rule: mean % of sign tokens recovered over all valid corpora in each cell. Score label: primary (no oracle sound correspondences). Orange: the plausible box. Lines: published sign-list sizes (top) and duplicate rates (right); solid lines are verified, dotted lines are read via a secondary source or come from an unverified preprint. Inventory is raised by allographs only, and no tested method merges them. The 2.9-4.2% range in the text is the box panel; this figure shows all valid corpora. Upper-bound maps: reports/sensitivity/appendix_sister_v1/.

4.5 Archaeological predictions (prediction, not measurement)

No seals-only, tablets-only or single-period corpus was tested; the regions below place qualitative published statements on sampler A's map. S, seals only, is the column with no duplicates ("almost all ... unique" seals; Kenoyer & Meadow 2010). T, tablets only, is the columns with duplicates 0.4-0.5, at or above the pooled M77 rate of 0.354, a derived lower bound. Both span inventories 400-800.

Table 9. Predicted regions, candidates tier.

Region	Primary, all corpora	Revised rule (post hoc)	Upper bound, with oracle sound correspondences
S, seals only	2.8-7.0%	6.5-24.4%	3.5-14.5%
T, tablets only	0.6-3.0%	2.5-11.7%	0.4-4.9%

S beats T at every inventory, under both rules. The strict panel (7 combinations valid in every cell) is less clear-cut: S leads only at 400, 500 and 800 signs, most at 400 (4.3% against 0.6%), and is within 0.4 points of T at 600 and 700. Under the frozen rule the seals-only advantage is concentrated at small inventories. Without a related language every region stays at or below 6.3%; the upper bound is 7.1%. A single-period corpus would use fewer signs than the pooled 400-450 (Kenoyer 2020b, p. 249), below anything we can build, so the map cannot place it. Every subset also holds fewer than 2,906 texts.

4.6 Robustness: a stricter sister

The stricter sister drops, for each corpus, every sister clause containing any hidden text's word sequence, so verbatim containment is zero by construction (Table 2 shows how much this removes; Sumerian keeps the most hidden word pairs, 19-22%). It covers seeds 0-2, both samplers, the Indus point and the equal-token comparison: 480 corpora.

Table 10. Corrected vs stricter sister, Indus point.

Quantity	A, corrected	A, stricter	B, corrected	B, stricter
related tier	15.9%	12.4%	30.2%	27.3%
candidates tier	2.4%	2.6%	12.9%	9.3%
candidates, with oracle sound correspondences	2.8%	2.6%	18.9%	12.3%
Original rule chose the sister (candidates)	20%	20%	33%	28%

Both headlines survive. The sampler contrast is 2.6% [1.2, 4.2] against 9.3% [5.4, 13.5], 3.7x, with non-overlapping intervals. Longer beats more by 13.6 [5.8, 21.3] points in sampler A and 24.1 [11.9, 35.9] in sampler B. One caution: the filter strips more of the sister from many short texts (40-52% of clauses) than from fewer long ones (9-15%), which favours the long-text condition, so the stricter sister shows that the length advantage survives without measuring its size better.

The solver picked the sister about as often as before, and related-tier scores fell by about 3 points. Most of what it recovers from the sister does not come from shared exact phrases.

4.7 Logo-syllabic corpora (most Indus-relevant script type)

"It is generally agreed that the signs found on seals and pottery represent a logosyllabic (morphemic) system" (Kenoyer 2020b, p. 249). Table 11 restricts every headline to logo-syllabic corpora (15 per point).

Table 11. Headlines for logo-syllabic corpora only.

Headline	Sampler A	Sampler B	Holds?
4.1 Entropy ratio, Rao top-100 merge (i.i.d. random control 0.696)	0.591 (fresh seeds 0.587)	-	Yes: the i.i.d. random control scores higher than the languages
4.2 No relative: mean, original / revised rule	0.0% / 0.8% (fresh seeds 0.1% / 0.9%)	0.3% / 1.1%	Yes: best single corpus 4.1%; none reaches 50%
4.3 Longer vs more at ~29k tokens, candidates	12.7% vs 0.2%	27.8% vs 6.4%	Yes
4.4 candidates, seeds 0-2	0.6% [0.0, 1.5]	3.6% [0.2, 9.3]	Not resolved: intervals overlap
4.4 candidates, seeds 3-5	0.9% [0.0, 2.6]	1.6% [0.2, 3.9]	Not resolved
4.4 candidates, stricter sister	1.1% [0.0, 3.2]	3.4% [0.2, 9.2]	Not resolved
4.4 related, seeds 0-2	3.6% [2.4, 4.7]	18.8% [6.6, 31.0]	Yes
4.4 related, seeds 3-5	3.2% [2.0, 4.4]	20.4% [8.3, 34.8]	Yes
4.4 related, with oracle sound correspondences	5.7%	35.6%	Yes

In the candidates tier logo-syllabic recovery is near zero for both samplers, and the reason is mundane. The original rule picks Sumerian for almost every logo-syllabic corpus and chooses the sister for only 3 of 15 in sampler A and 4 of 15 in sampler B, mostly when the hidden language is Sumerian itself. The sampler contrast shows only when the sister is the sole reference.

4.8 Genre check: Sumerian seal inscriptions only

Four of five sources are literary. As a check, we built Sumerian hidden corpora and sisters only from the 19,076 Ur III seal inscriptions in the CDLI dump (Indus point, seeds 0-2, four script types, 12 corpora per sampler) and compared them with full-source Sumerian.

Table 12. Sumerian, seal-only vs full source.

Tier and score	A, seal only	A, full source	B, seal only	B, full source
related	13.1%	5.0%	8.8%	13.7%
candidates	9.0%	5.0%	7.7%	13.7%
candidates, with oracle sound correspondences	26.1%	6.8%	22.8%	17.2%
none, original rule	1.5%	2.2%	3.6%	2.3%

Seal-only recovery is of the same order as full-source recovery, with no consistent direction. The no-relative tier stays low: means at most 3.6%, best corpus 20.2%. The oracle step adds more on seal-only material, 15-17 points. The check is weak: seal texts are far more repetitive and use fewer sign types (Table 13), and none meets every calibration target, since the knobs were calibrated on the full source. It is consistent with

the main conclusions not depending on literary genre for Sumerian, but it is not a calibrated Indus-point result and cannot settle the question.

Table 13. Seal-only Sumerian corpora against the calibration targets.

Statistic	Seal-only corpora	Target
Duplicate-text rate	0.51-0.59	not a target
Sign types	about 340	400-700
Signs covering 80% of tokens	37	69

4.9 Positive controls

EM on English letter-substitution ciphers. Knight et al. (2006, Sec. 3) decipher a 417-letter encyclopedia article under a 1:1 letter substitution, with an English letter-bigram model and known word boundaries. Their bigram EM/Viterbi baseline gets 83.7% of letters right (68 errors); their best configuration reaches 97.6%. We ran the frozen EM at benchmark settings on the same kind of task: plaintext models from *Pride and Prejudice* and *Moby Dick* at their two data sizes, ciphertext from *On the Origin of Species*, word boundaries given, 10 random passages and keys per length.

Table 14. Frozen EM on English letter-substitution ciphers: letters decoded correctly.

Cipher length (letters)	100	200	417	1,000	2,000	5,000
Original rule, 1.5M-character model	58.2%	92.2%	97.5%	99.7%	99.9%	99.9%
Original rule, 70,000-character model	71.7%	84.6%	97.4%	99.8%	99.9%	99.9%
Revised rule, 1.5M-character model	60.9%	92.7%	97.7%	99.6%	99.7%	99.8%

At the published length of 417 letters the frozen EM decodes 97.4-97.7% of letters: above the published baseline, level with the best configuration. The settings differ in detail (Viterbi over the whole text vs one value per cipher letter; novels and Darwin vs news and an encyclopedia), but the solver plainly works on the problem it was built for. Its low recovery on Indus-like corpora is not a broken implementation.

Conditional-entropy ordering (Rao et al. 2009). Rao et al. give these values only as figures, which we read by eye; the supplement tabulates only Indus perplexities. We applied the frozen entropy function to our own reference corpora at matching unit levels, with a shuffled sequence as the random reference.

Table 15. Relative conditional entropy: Rao et al. (2009) vs our corpora.

Corpus	Rao et al., Fig. 1B (by eye)	Ours, relative, 400 tokens	Ours, nats, 400 tokens
i.i.d. random control (their type 1)	1.00	1.00	5.45
Sanskrit	0.66	0.75	3.35
English words	0.64	0.81	3.31
Sumerian	0.57	0.67	3.24
Old Tamil	0.56	0.74	3.34
English characters	0.51	0.85	2.47
Rigid control (their type 2)	about 0	0.02	0.08

The extremes reproduce, and at 400 tokens our languages sit at 3.2-3.4 nats, close to the roughly 3.3-3.8 of Rao's Fig. 1A. The ordering among languages does not: English characters rank highest for us and lowest for Rao, and Sanskrit leads Sumerian in both, by different margins. Our corpora (Ramayana, CDLI and Pride and Prejudice rather than Rig Veda, ETCSL and the Brown corpus), unit definitions and random reference all differ, and English characters have fewer than 400 types. This is a partial reproduction only.

5. Limitations

1. **Allograph merging.** Inventory rises only through allographs, which no tested method merges.
2. **Sister language.** The sister keeps word order and morphology, shares source text with the hidden corpus (verbatim overlap 30-56% by language, corrected sister) and comes with oracle sound correspondences. Its distance is a free parameter. Sister tiers are optimistic.
3. **Oracle script type and unit level in Task D.** Real decipherers are not told the script type.
4. **Positive controls are partial.** The frozen EM matches the best published cipher accuracy; the entropy statistic reproduces Rao et al.'s extremes and magnitude but not their ordering. The cognate-matcher and the n-gram, positional, segmentation and script-type methods have no positive control, so a low score from them could still reflect our implementation.
5. **Genre.** Four of five sources are literary (an epic, classical poetry, Caesar and Vergil, an oral epic); only Sumerian includes real seal inscriptions, while Indus texts are mostly seals and tablets. The Sumerian seal-only check agrees in order of magnitude but misses the calibration targets.
6. **One language, one fixed script per corpus.** The Indus script changed over about 700 years (Kenoyer & Meadow 2010, p. 8) and may have written more than one language (Kenoyer 2020b, pp. 237, 240, 254). A corpus mixing languages or script stages is not modelled, and could be harder than anything tested here.
7. **Few clusters.** With five source languages, the percentile cluster bootstrap can give intervals that are too narrow; coverage was not measured, so all intervals are approximate. Logo-syllabic subsets have only 15 corpora per point.
8. **Text-beginner calibration.** Most corpora miss this target (32-37% meet it, corrected-sister runs), and sampler B did not fix it.
9. **Median length is derived, not published.** The target of 4 comes from Yadav et al. 2010, Fig. 2, which is not like-for-like with our corpora (Table 1), and was originally unsourced. It also sets the spread of generated lengths, through $\sigma = (\ln 17 - \ln 4) / z$. Dropping it from the calibration objective would change nothing, since every corpus meets it and only unmet targets count; dropping it from the sampler would need another assumption for the spread, with effects unknown without rerunning.
10. **CDLI licence.** Academic reuse with citation, not open; Sumerian stays out of published rounds.
11. **The 2,591 vs 2,906 discrepancy** in the source of the M77 duplicate rate and the median.
12. **Subset positions are estimated** from qualitative statements and raw counts.

13. **The single-period subset cannot be placed**; every subset is smaller than 2,906 texts.
14. **Post-hoc decisions**. The revised rule, sampler B, the sister corrections and several analyses followed results; Appendix A lists each. The original rule stays primary.
15. **Plug-in entropy is biased at this scale**, so fixed entropy cut-offs depend on corpus size.
16. **Controls built by us**. A classifier is only as general as its controls.
17. **No neural decipherment model is evaluated**. The EM cognate-matcher contains no neural network.

6. Blind challenge and reproducibility

Public challenge rounds release corpora and keys; hidden rounds release corpora and SHA-256 commitments, with keys held by the maintainer outside the repository. Challenge corpora come from secretly disguised sister languages, so they cannot be matched against public source texts. Each team gets one scored submission per hidden round, Sumerian-derived corpora are excluded, and the static leaderboard marks entries that claim a real Indus decipherment.

Every number here can be regenerated from the repository. Method code is frozen at the git tag `frozen-v1` and the data release at `data-v1`; later changes touch only data generation and analysis, selected per run profile, and each run records its sampler and sister version. Internally, samplers A and B are generator `v1` and `v2`, and the original, corrected and stricter sisters are `sister-v1`, `sister-v2` and `sister-v3`. Source texts are fetched by the user under their own licences and never redistributed.

7. Conclusion

[AUTHOR REWRITE. Current draft text, kept for reference:]

On synthetic corpora that match the published Indus statistics, mean recovery without a related language stays at or below 4.8% in every run, and longer inscriptions help more than more inscriptions. Published Indus statistics do not fix how decipherable an Indus-scale corpus is: two samplers that the published statistics cannot tell apart differ roughly three- to fivefold when a related language is available, and for logo-syllabic corpora, the most Indus-relevant type, this contrast appears only when the related language is the sole reference; among candidate languages it is unresolved. A statement about how hard the Indus script is to decipher therefore has to name its corpus (object types, period, duplication), its knowledge tier, its script type and its selection rule. None of these results deciphers the Indus script or says anything about what language it records, or whether it records a language at all.

Appendix A. Analysis history

Every decision taken after any result was seen, in order, with the commit that records it. Decisions taken before any result are marked "pre-results". Dates are 2026, UTC.

#	Date	Decision	Pre-results or post-results	Commit
1	3 Oct	Design review fixes six safeguards: band and feasibility reporting, three calibration regimes, structural controls, knowledge tiers, "no method succeeds" regions reported, probe-resistant challenge	Pre-results	b1194b0
2	3 Oct	Calibration targets corrected from unverified initial planning values (5,500 texts, mean 4.4) to Mahadevan (1977) via Yadav et al. 2010 and Rao 2018 (2,906 texts, mean 4.60); all results regenerated. The median target (4) was kept from the initial planning values, unsourced	Post-results (development runs)	05b91d3
3	3 Oct	Two EM corrections during development (pooled rare-unit state pinned; candidate selection by normalised bigram gain) produce the revised rule. The rule as first written is kept as the original rule and both are reported	Post-results (development runs)	e625f56
4	3 Oct	Method code frozen at frozen-v1; Sumerian excluded from published challenge rounds pending CDLI terms	Pre-final-run	e625f56
5	3 Oct	Final run, 1,554 corpora	-	b1f8704
6	3 Oct	Length sweep added to the replication profile; fresh-seed replication (seeds 3-5) run	Post-results	bbd0e49, d997015
7	3 Oct	Generator v2 written to fix the missed text-beginner target (did not fix it)	Post-results	7246062, fde7031
8	3 Oct	Sensitivity sweep over duplicate rate and inventory added; headline order changed to lead with generator dependence	Post-results	4dcb9a1
9	3 Oct	Plausible box (duplicates 0.2-0.4, inventory 400-700) written into the profile after a test rendering of the first 24 sweep records and a reachability count on 1,216 records, before sweep recovery was tabulated. First committed together with the sweep results, so the order rests on the session log, not on git	Partly pre-results (see text)	a0e1233
10	3 Oct	M77 duplicate rate (0.354) measured from the vector data of Yadav et al. 2010, Fig. 2, and marked on the map as a reference point	Post-results	a0e1233
11	4 Oct	After input from J. M. Kenoyer: "stray finds" and "secure context" wording removed; object-type and period notes added; archaeological subsets placed on the map as predictions	Post-results	00f243f, 30a7ec5
12	4 Oct	Original vs revised rule reanalysed (no new runs). The original rule made primary throughout; "original rule stays near 5% in the box" found to hold for generator v1 only	Post-results	8007ed7
13	4 Oct	Audit of the 93% corpus finds no key leak or scoring bug, but a sister clause-split defect and the size of the oracle cognate step	Post-results	900d104
14	4 Oct	Sister-v2 (split by clause content) and before-cognate scoring; sister-v1 numbers relabelled "upper bound (sister-v1, known overlap)"; primary redefined as original rule before the cognate step	Post-results	0a5c913, 87d5706
15	5 Oct	Sister-v3 robustness check and equal-token size points	Post-results	1cb6faf
16	5 Oct	Headline 3 wording changed from "roughly fivefold" to "roughly three- to fivefold" after replication (3.5x) and sister-v3 (3.7x)	Post-results	457be13
17	5 Oct	Residual-overlap figure corrected from 38-82% (one script type and seed, estimated under the old split) to 30-56% (12 corpora	Post-results	40acfba

#	Date	Decision	Pre-results or post-results	Commit
		per language, measured under sister-v2)		
18	5 Oct	Median target relabelled, first as "unsourced", then as "derived from Yadav et al. 2010, Fig. 2"; it was originally taken from the initial planning values without a source	Post-results	40acfba, this revision
19	5 Oct	Sensitivity sweep rerun under sister-v2 with before-cognate scores	Post-results	this revision
20	5 Oct	Logo-syllabic headline split and Sumerian seal-only genre check added	Post-results	this revision
21	5 Oct	One configuration typo caused 60 failed jobs in an equal-token profile; fixed and rerun, failed records discarded	Operational	1cb6faf
22	5 Oct	Headlines reordered (length vs size becomes headline 3; underdetermination becomes headline 4) and headline 4 reworded to state the logo-syllabic result	Post-results	this revision
23	5 Oct	Positive controls added: frozen EM on English letter-substitution ciphers, and conditional-entropy ordering against Rao et al. (2009) figures. The latter showed that our control keys number Rao's type 1 and type 2 the other way round	Post-results	this revision
24	5 Oct	Sensitivity sweep aggregated with before-cognate scores. Seals-only and tablets-only regions defined as duplicates 0.0 and 0.4-0.5 at all inventories, for both sweeps; an earlier tablets-only range (1.4-4.9%) had used the 0.4 column only	Post-results	this revision
25	5 Oct	Nair (2026) availability checked: on request only, no public repository or licence	-	this revision

Back matter

Data and code availability. Code, configurations and run records are available from the author on request.

Acknowledgements. Code was developed with AI coding assistance.

Appendices. Analysis history (Appendix A); sweep maps and tables, primary (reports/sensitivity/) and upper bound (reports/sensitivity/appendix_sister_v1/); calibration, full result tables, sensitivity tables, the rule comparison, the audit, the original, corrected and stricter sister comparisons, and method deviations.

References

Author-year style. Only sources cited in the text. "(not verified)" marks an entry, or the part of it in brackets, that we have not checked against the source itself.

Berg-Kirkpatrick, T., & Klein, D. (2013). Decipherment with a million random restarts. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing* (pp. 874-878).

Chao, A. (1984). Nonparametric estimation of the number of classes in a population. *Scandinavian Journal of Statistics*, 11, 265-270. (not verified)

Farmer, S., Sproat, R., & Witzel, M. (2004). The collapse of the Indus-script thesis: The myth of a literate Harappan civilization. *Electronic Journal of Vedic Studies*, 11(2) [pp. 19-57,

not verified].

Harris, Z. S. (1955). From phoneme to morpheme. *Language*, 31(2), 190-222. (not verified)

Kenoyer, J. M. (2020a). The origin and development of the Indus script: Insights from Harappa and other sites. In K. Lashari (Ed.), *Studies on Indus Script* (pp. 217-236). Karachi: National Fund for Mohenjodaro.

Kenoyer, J. M. (2020b). The Indus script: Origins, use and disappearance. In H. Zhao (Ed.), *Dialogue of Civilisation: Comparing Multiple Centers* (pp. 220-255). Shanghai: Shanghai Guji Press. [Page numbers cited are those printed in the pre-publication PDF.]

Kenoyer, J. M., & Meadow, R. H. (2010). Inscribed objects from Harappa excavations 1986-2007. In A. Parpola, B. M. Pande & P. Koskikallio (Eds.), *Corpus of Indus Seals and Inscriptions, Vol. 3: New Material, Untraced Objects, and Collections outside India and Pakistan, Part 1* (Annales Academiae Scientiarum Fennicae, Humaniora 359). Helsinki: Suomalainen Tiedeakatemia. [Page numbers cited are those of the PDF read.]

Knight, K., Nair, A., Rathod, N., & Yamada, K. (2006). Unsupervised analysis for decipherment problems. In *Proceedings of the COLING/ACL 2006 Main Conference Poster Sessions* (pp. 499-506).

Lee, R., Jonathan, P., & Ziman, P. (2010). Pictish symbols revealed as a written language through application of Shannon entropy. *Proceedings of the Royal Society A*. doi:10.1098/rspa.2010.0041 [466(2121), 2545-2560, not verified].

Luo, J., Cao, Y., & Barzilay, R. (2019). Neural decipherment via minimum-cost flow: From Ugaritic to Linear B. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics* (pp. 3146-3155).

Mahadevan, I. (1977). *The Indus Script: Texts, Concordance and Tables* (Memoirs of the Archaeological Survey of India 77). New Delhi: Archaeological Survey of India. (not verified; figures read via Yadav et al. 2010 and Rao 2018)

Nair, A. (2026). How non-linguistic is the Indus sign system? A synthetic-baseline scorecard. arXiv:2604.17828 (preprint, not peer reviewed).

Parpola, A. (1994). *Deciphering the Indus Script*. Cambridge: Cambridge University Press. (not verified; sign count read via Rao 2018)

Possehl, G. L. (2002). *The Indus Civilization: A Contemporary Perspective*. Walnut Creek, CA: AltaMira. (not verified; figures read via a secondary source)

Rao, R. P. N. (2018). The Indus script and economics. In *Walking with the Unicorn: Social Organization and Material Culture in Ancient South Asia* (pp. 518-525). Oxford: Archaeopress. arXiv:1812.00049.

Rao, R. P. N., Yadav, N., Vahia, M. N., Joglekar, H., Adhikari, R., & Mahadevan, I. (2009). Entropic evidence for linguistic structure in the Indus script. *Science*, 324, 1165.

Snyder, B., Barzilay, R., & Knight, K. (2010). A statistical model for lost language decipherment. In *Proceedings of the 48th Annual Meeting of the Association for Computational Linguistics* (pp. 1048-1057).

Sproat, R. (2010). Last words: Ancient symbols, computational linguistics, and the reviewing practices of the general science journals. *Computational Linguistics*, 36(3), 585-

594. doi:10.1162/coli_a_00011

Sproat, R. (2014). A statistical comparison of written language and nonlinguistic symbol systems. *Language*, 90(2), 457-481.

Tanaka-Ishii, K. (2005). Entropy as an indicator of context boundaries: An experiment using a web search engine. In *Proceedings of the Second International Joint Conference on Natural Language Processing (IJCNLP 2005)* [pp. 93-105, not verified].

Tiwari, T. (2026). Statistical structure in Indus sign sequences. In *Proceedings of the 6th International Conference on Natural Language Processing for the Digital Humanities* (pp. 314-319).

Wells, B. K. (2006). *Epigraphic Approaches to Indus Writing* (PhD thesis). Harvard University. (not verified; sign count read via Yadav et al. 2010)

Wells, B. K. (2015). *The Archaeology and Epigraphy of Indus Writing*. Oxford: Archaeopress. (sign count not verified; read via Rao 2018)

Yadav, N., Joglekar, H., Rao, R. P. N., Vahia, M. N., Adhikari, R., & Mahadevan, I. (2010). Statistical analysis of the Indus script using n-grams. *PLoS ONE*. arXiv:0901.3017.

Data sources. Cuneiform Digital Library Initiative (CDLI), bulk ATF data dump, August 2022, <https://cdli.earth> (reused under CDLI terms: academic reuse with citation). Digital Corpus of Sanskrit, O. Hellwig, <https://github.com/OliverHellwig/sanskrit> (CC BY 4.0). Project Madurai, <https://www.projectmadurai.org>. Project Gutenberg, <https://www.gutenberg.org> (texts 218, 229, 231, 1228, 1342, 2701, 7000).